Draft version October 22, 2025
Typeset using LaTeX **RNAAS** style in AASTeX631

Applications of the Roots of Bessel Functions of the First Kind

Madeline Renee Boss[1] and Samhitha Devi Kunadharaju[1]

[1] *University of Texas at Austin*

ABSTRACT

This paper investigates the physical applications and numerical computations of roots for Bessel functions of the first kind. Using Python's `scipy` library and the `fsolve` function, we compute the first five positive roots for $n = \{0, 1, 2\}$, obtaining $J_0 = \{2.40, 5.52, 8.65, 11.79, 14.93\}$, $J_1 = \{3.83, 7.02, 10.17, 13.32, 16.47\}$, and $J_2 = \{5.13, 8.42, 11.62, 14.80, 17.96\}$. We examine how these roots determine the physical behavior in three applications: the mode patterns of vibrations of a circular drumhead, the disappearance of sidebars in frequency modulation (FM) signal synthesis, and the suppression of certain harmonic components of Kepler's equation for planetary motion. This work demonstrates the fundamental role of Bessel function roots in problems across various fields of physics.

Keywords: Bessel Function, Roots, Circular Vibrations, Frequency Modulation (FM), Kepler's Equation

INTRODUCTION

The Bessel differential equation is a second-order linear differential equation expressed as:

$$x^2y'' + xy' + (x^2 - n^2)y = 0 \tag{1}$$

where x is a real or complex variable, $y(x)$ is the dependent variable, y' and y'' denote the first and second derivatives of y with respect to x, and n (an integer or real number) is called the *order* of the Bessel function.

The solutions to this equation, known as Bessel functions of the first kind, are denoted by $J_n(x)$ and are given by the integral representation:

$$J_n(x) = \frac{1}{\pi}\int_0^{\pi} \cos(x\sin\theta - n\theta)\, d\theta \tag{2}$$

for integer orders $n = 0, 1, 2, \ldots$

These functions arise naturally in physical problems that exhibit cylindrical or radial symmetry, including heat conduction, wave propagation, and vibrational motion.

The equation was first systematically analyzed by Friedrich Wilhelm Bessel in 1824 during his work on celestial mechanics, particularly in solving Kepler's equation for planetary motion. However, its form had already appeared a century earlier in the works of the Bernoulli brothers, Euler, and Fourier, who encountered special cases while studying oscillations and heat flow (Dutka 1995).

As Dutka notes, these early investigations demonstrated how problems in mechanics and astronomy—especially oscillations of strings and orbital motion—naturally led to what are now recognized as Bessel functions.

In this paper, we numerically determine the first five roots of $J_n(x)$ for $n = 0, 1, 2$ with $x = [0,\ 20]$ and interpret their roles in three physical scenarios:

1. Vibration of a circular drumhead
2. Frequency modulation (FM) signal synthesis

Corresponding author: Madeline Renee Boss, Samhitha Devi Kunadharaju
madelinereneeboss@gmail.com, samhitha.dk@gmail.com

3. The solution of Kepler's equation for planetary motion

APPLICATIONS OF THE BESSEL FUNCTION

With the background of the Bessel function covered, the rest of this paper will focus on its practical applications. One of the simplest situations in which the Bessel function can be used is determining vibration modes for a circular drumhead, a problem first investigated by Leonhard Euler Dutka 1995. The equation used to solve vibration modes is:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u. \tag{3}$$

Another usage is in frequency modulation (FM) synthesis, which uses the equation

$$s(t) = A_c \cos(\omega_c t + \beta \sin(\omega_m t)). \tag{4}$$

The application of the Bessel function in this scenario arises from the Jacobi–Anger expansion of the Euler form of the FM equation, which introduces Bessel functions of the first kind into the expression.

Finally, Bessel functions also appear in celestial mechanics, particularly as part of the solution to Kepler's equation, which relates a planet's mean anomaly (M), eccentric anomaly (E), and orbital eccentricity (e) through the relation

$$M = E - e \sin E. \tag{5}$$

Here, the mean anomaly represents the average angular position of a planet in its orbit over time, the eccentric anomaly describes its actual angular position assuming a circular reference orbit, and the orbital eccentricity quantifies how stretched or non-circular the orbit is. Using a Fourier-series expansion involving Bessel functions, E can be expressed as

$$E = M + 2\sum_{n=1}^{\infty} \frac{1}{n} J_n(ne) \sin(nM), \tag{6}$$

allowing periodic orbital motion to be computed accurately even for eccentric orbits (Dutka 1995).

FINDING ROOTS FOR BESSEL FUNCTION

To find the first five roots of the Bessel function for $n = \{0, 1, 2\}$, a Python script was developed to estimate the roots' exact positions. To achieve this, the script first created a function for the integral part of the Bessel equation, which took in values of θ, n, and x. Next, a function was created that combined the integral part of the Bessel function with its other components, taking in the values of n and x.

To combine the integral portion, the function `quad` from the `scipy` library was used to calculate the integral from $\theta = 0$ to π for the given values of n and x. Afterwards, it calculated the rest of the Bessel function for those same inputs. Figure 1 shows a visualization of the Bessel functions for $n = \{0, 1, 2\}$ over the range $x \in [0, 20]$.

After creating a Python version of the function, the `scipy` library's `fsolve` function was used to estimate the exact positions of the roots. The `fsolve` method estimates the root of a nonlinear equation by iteratively approximating the Jacobian matrix—the matrix of partial derivatives—until the solution converges within an error tolerance of 10^{-6}. Figure 2 shows the computed root points found by the `fsolve` function.

The roots for the Bessel functions are as follows:

- For $n = 0$: $\{2.40, 5.52, 8.65, 11.79, 14.93\}$
- For $n = 1$: $\{3.83, 7.02, 10.17, 13.32, 16.47\}$
- For $n = 2$: $\{5.13, 8.42, 11.62, 14.80, 17.96\}$

APPLICATION OF ROOTS

With the roots discovered, we can now examine what those roots mean in the context of the three examples of the Bessel function discussed earlier. Starting with vibrations of circular drumheads, the Bessel function roots can be used

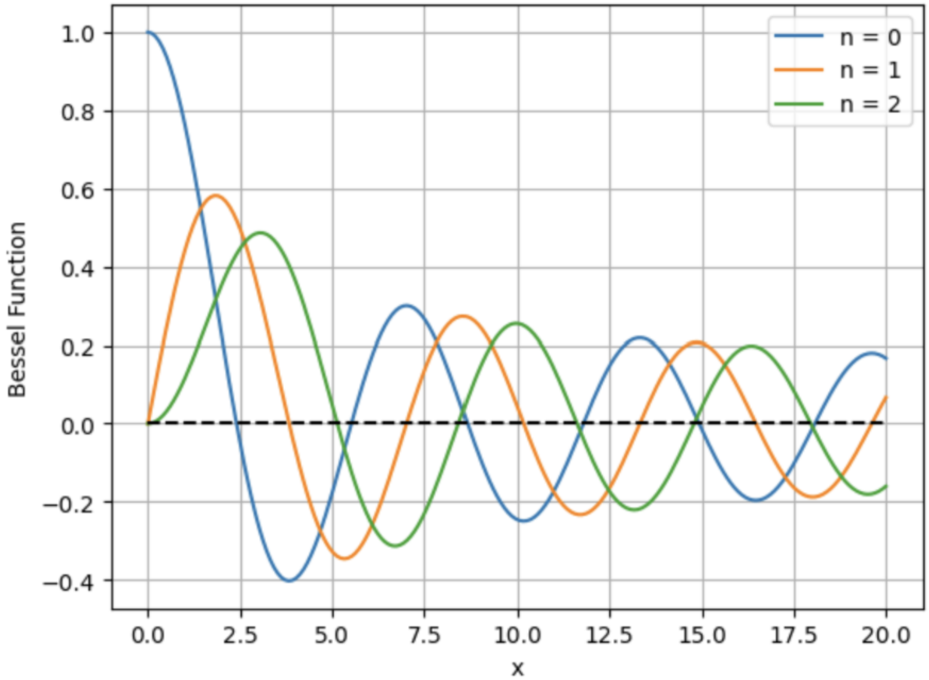

Figure 1. Visualization of Bessel functions $J_0(x)$, $J_1(x)$, and $J_2(x)$ for $x \in [0, 20]$.

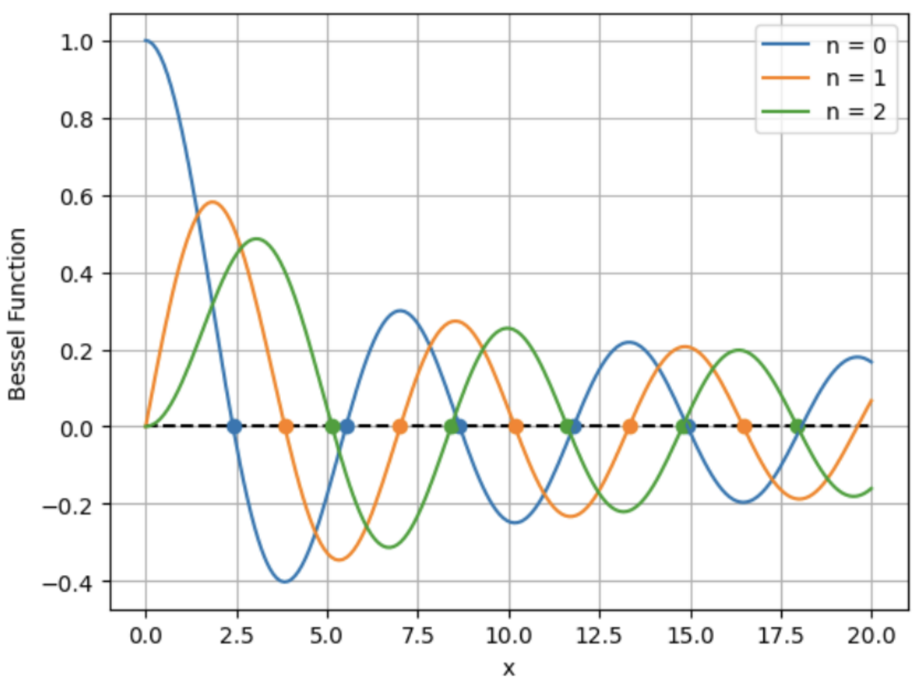

Figure 2. Estimated roots of Bessel functions $J_0(x)$, $J_1(x)$, and $J_2(x)$ obtained using `fsolve`.

to classify different vibration patterns. When determining the displacement of a circular drumhead, the governing equation is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u. \tag{7}$$

However, when analyzing the radial displacement (the displacement at the edges of the drum), this equation can be reduced to

$$r^2 R'' + rR' + (\lambda^2 r^2 - m^2)R = 0. \tag{8}$$

The boundary condition at the edge of the drum requires the displacement to be zero, setting the equation to equal zero at the rim. Since this equation now takes the same differential form as the Bessel function, we can use its roots to satisfy the condition for values of λ, with m representing different orders n. For each combination of λ and m, the vibration patterns change, creating unique modes of oscillation. Figure 3 shows visualizations of these vibrational modes, created by L. V. Jindong and Akshat Vijoy for their paper on the effects of air pressure on circular vibrations (Jindong and Vijoy 2023).

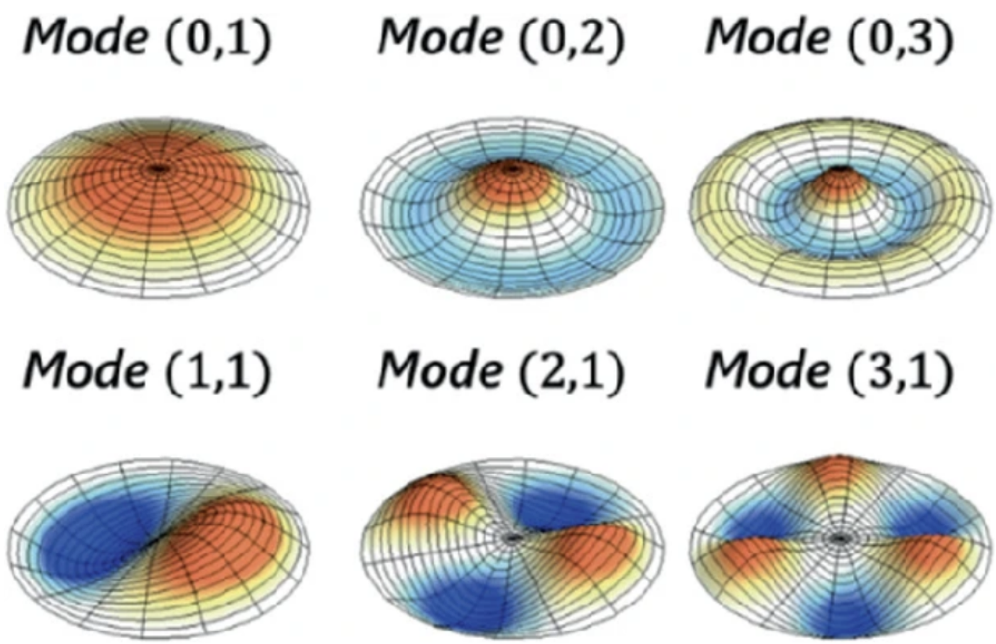

Figure 3. Vibration modes of a circular drumhead corresponding to different Bessel function roots. Image adapted from L. V. Jindong & Akshat Vijoy (Effects of Air Pressure on Circular Vibrations).

Next, we can examine the role of Bessel roots in frequency modulation (FM) synthesis. As described by Thomas and Sekhar in *Communication Theory* (Thomas and Sekhar 2005), the Bessel function within the FM equation is primarily used to determine the sideband amplitude of a sine wave for a given modulation index (x) and sideband number (n). The following equation is the Bessel function:

$$s(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \cos\big((\omega_c + n\omega_m)t\big) \tag{9}$$

And the expansion for $J_n(\beta)$ is:

$$x^2 \frac{d^2 y}{dx^2} + x\frac{dy}{dx} + \left(x^2 - n^2\right)y = 0 \tag{10}$$

When a $J_n(\beta)$ reaches zero the associating sideband's power is redistributed among the remaining sidebands. For instance, when the modulation index reaches approximately $x = 2.40$, the power of the carrier component ($n = 0$)

vanishes, causing the sidebands to dominate. This property is exploited in FM systems to reduce interference or emphasize specific harmonics (Thomas and Sekhar 2005).

Finally, in the case of Kepler's equation, the roots of the Bessel function help interpret how the periodic components of a planet's orbit vary with orbital eccentricity. Kepler's equation,

$$M = E - e \sin E, \tag{11}$$

can be expressed as a Fourier series involving Bessel functions of the first kind:

$$E = M + 2 \sum_{n=1}^{\infty} \frac{1}{n} J_n(ne) \sin(nM). \tag{12}$$

In this representation, each term $J_n(ne)$ defines the amplitude of the nth harmonic in the planet's orbital motion. The roots of $J_n(x)$ indicate specific values of the eccentricity e at which the corresponding harmonic vanishes, meaning that certain periodic components contribute no net effect to the planet's position or velocity. This provides a natural means of identifying resonance conditions or simplified orbital configurations. When $J_n(ne) = 0$, that harmonic's influence disappears, and the orbital motion is dominated by the remaining nonzero harmonics.

As discussed in *On the Bessel Solution of Kepler's Equation* (Borghi 2024), the series expansion using Bessel functions offers both analytical precision and physical interpretability, revealing transitions between different harmonic regimes of orbital motion. Thus, the roots of the Bessel function serve as natural markers for changes in orbital dynamics, similar to how they define vibration modes in a drumhead or determine carrier suppression in FM synthesis.

FUTURE WORK

Overall, the numerical methods used in this project, SciPy's integration and root-finding functions, proved effective for approximating and visualizing the Bessel functions of the first kind. Based of the plot, the computed roots closely aligned with the theoretical values.

While this project focused on computing and visualizing the first five roots of the Bessel functions, there are several directions that could deepen this analysis. One potential next step would be to examine higher-order Bessel functions or the modified Bessel functions $I_n(x)$ and $K_n(x)$, which arise in problems involving heat transfer and electromagnetic waves.

Additionally, we could numerically approximate Bessel roots for complex or non-integer orders. These functions appear in more advanced applications such as quantum mechanics, where symmetry is not purely circular.

Finally, it would be interesting to connect the theoretical roots more directly to real-world systems, such as analyzing how small perturbations in eccentricity affect the Bessel-based solution of Kepler's equation, or how the spacing of roots relates to resonance patterns in acoustics.

Any of these directions would provide a more complete understanding of how Bessel functions underpin a wide range of physical phenomena.

ACKNOWLEDGMENTS

We appreciate the guidance and support provided in the course *CS323E Elements of Scientific Computing*. We thank our teaching assistant, Omatharv Vaidya, for promptly answering questions and offering helpful clarifications online, and we thank Dr. Shyamal Mitra for his instruction and overall direction throughout the semester.

REFERENCES

Borghi, R. (2024). On the bessel solution of kepler's equation. *Mathematics*, 12(1):154.

Dutka, J. (1995). On the early history of bessel functions. *Archive for History of Exact Sciences*, 49(2):105–134.

Jindong, L. V. and Vijoy, A. (2023). Quantifying the effects of air pressure on a vibrating tensioned elastic membrane. In *Proceedings of the 9th IRC Conference on Science, Engineering, and Technology*, pages 405–415.

Thomas, T. G. and Sekhar, S. C. (2005). *Communication Theory*. Tata-McGraw Hill.